Smart stock 2026-09-11

⚠ We couldn't load the file. SmartStock – System Architecture Document

Version	Date	Author	Description of Changes
A	Sep 10, 2026	Dan Uziel Aquino	Initial Document
B	Sep 18, 2026	Dan Uziel Aquino	Aligned database engine with Infrastructure and Development Workflow docs (PostgreSQL via Supabase, not MySQL/SQL Server)

High-Level System Architecture

SmartStock is a web-based inventory management system designed to help businesses monitor and manage their products and stock levels. The system uses a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

The Presentation Layer provides the interface for administrators and authorized staff. The Application Layer handles inventory operations, user authentication, stock monitoring, and reporting. The Database Layer stores product information, stock records, users, transactions, and reports.

Detailed System Architecture

The SmartStock system consists of the following major components:

1. User Interface – Provides dashboards and pages for administrators and staff.
2. User Management – Manages user accounts, roles, and access permissions.
3. Product Management – Allows users to add, edit, delete, and view product information.
4. Inventory Management – Monitors available stock and inventory levels.
5. Stock-In Management – Records newly received products and updates inventory quantities.
6. Stock-Out Management – Records released or sold products and automatically updates stock levels.
7. Low Stock Monitoring – Identifies products that have reached their minimum stock level.
8. Supplier Management – Stores and manages supplier information.
9. Reports – Generates inventory, stock-in, stock-out, and low-stock reports.
10. Database – Stores all SmartStock system records securely.

Technical Approach

SmartStock will be developed as a web-based inventory management system that can be accessed through a web browser. The front end will provide a simple and user-friendly interface for managing inventory.

The back end will handle business logic, user authentication, product management, stock transactions, and report generation. **PostgreSQL (hosted on Supabase)** will be used to store product, user, supplier, and transaction records — consistent with the Infrastructure and Development Workflow documentation.

The system will use a modular approach where each inventory function is separated into its own module to make the system easier to maintain and update.

> **Note:** Conceptual diagram images from the original draft were not carried over in this edit (they were embedded as local `blob:` references that don't render for other viewers). Please re-attach the diagrams as uploaded images before Saturday's review.

Solution Assumptions / Constraints / Dependencies

Assumptions

- Users have access to a computer or mobile device with a web browser.
- Users have valid SmartStock accounts.
- Users are assigned appropriate roles and permissions.
- Product and inventory information will be provided by authorized personnel.
- Staff will update stock transactions regularly.

Constraints

- The system requires an internet or local network connection depending on deployment.
- Only authorized users can access inventory information and management functions.
- System performance depends on available server, database, and network resources.
- Incorrect stock entries may result in inaccurate inventory records.

Dependencies

- Web browser
- Web server (Render)
- PostgreSQL database server (Supabase)
- Internet or local network connection
- Authentication and security components

Architectural Decisions

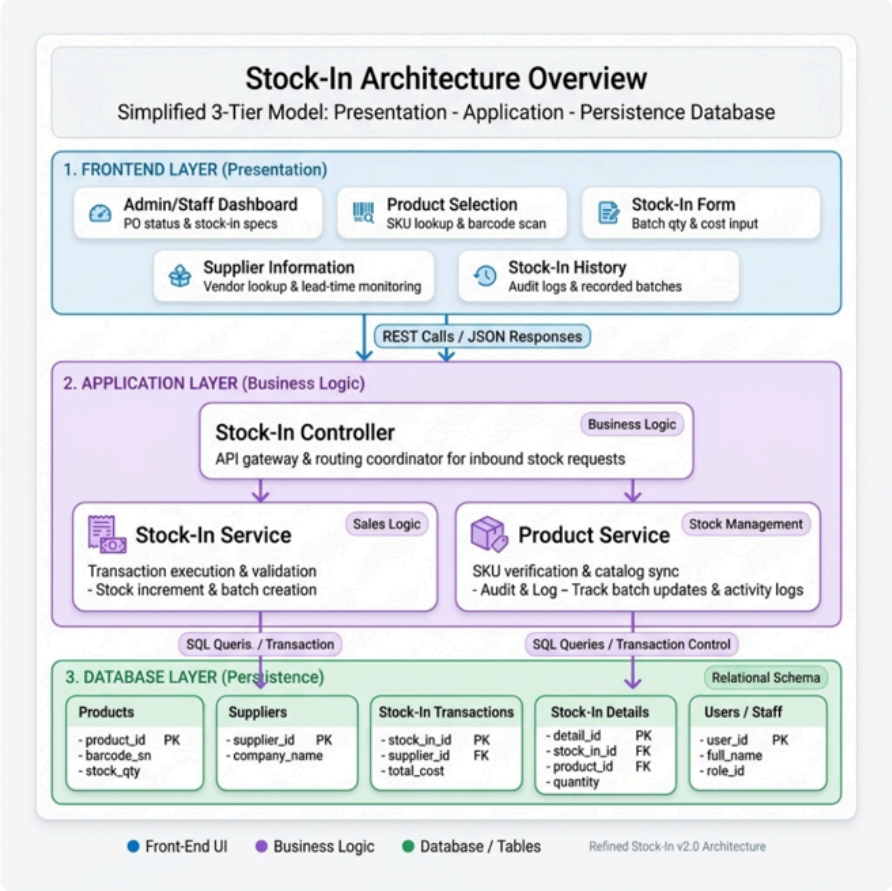
1. **Three-Tier Architecture** – Used to separate the user interface, business logic, and database for easier maintenance.
2. **Web-Based System** – Selected so users can access SmartStock through a web browser.
3. **Role-Based Access Control** – Used to restrict system functions based on the user's role.
4. **Relational Database (PostgreSQL via Supabase)** – Used to properly organize products, suppliers, users, and inventory transactions.
5. **Modular Design** – Used to separate inventory functions into manageable modules.
6. **Automated Stock Updating** – The system will automatically update product quantities whenever stock-in or stock-out transactions are recorded.
7. **Low-Stock Notification** – The system will identify products with low quantities so staff can restock them promptly.

Admin/Staff – Stock-In Management

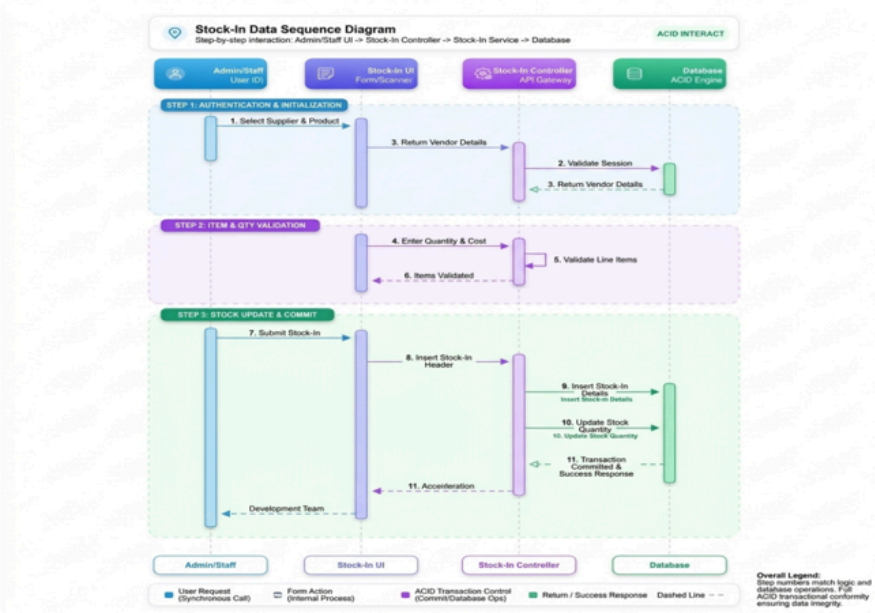
Feature Description

The Stock-In Management feature allows Admin/Staff to record products received from suppliers. The system records the product, quantity, unit of measure, supplier, staff member, and date and time. After saving the transaction, the system automatically increases the inventory quantity. It also supports unit conversion for bulk products such as boxes of screws.

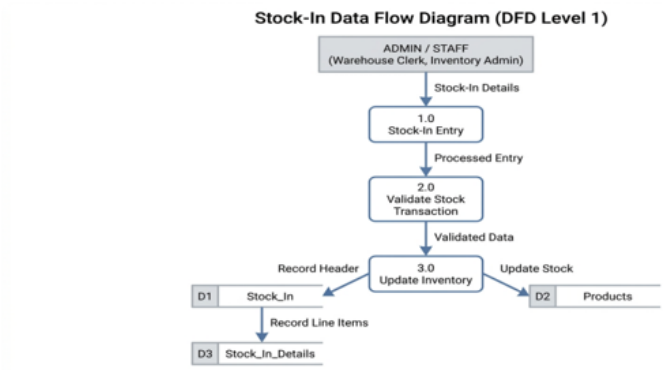
Low-level detailed system / arch design



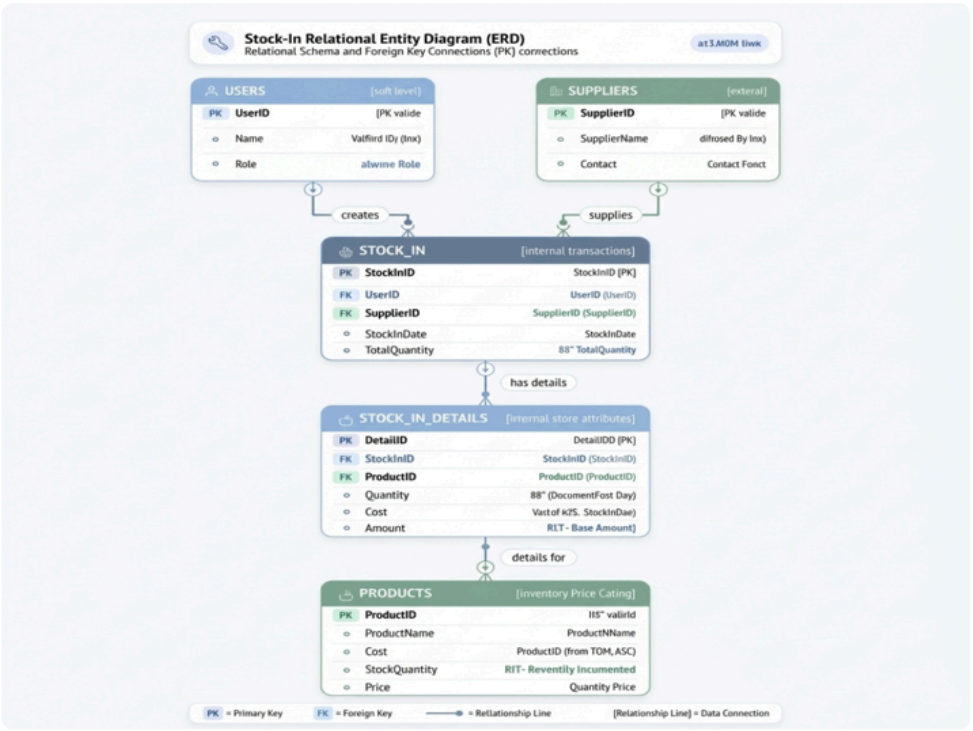
Sequence Diagram



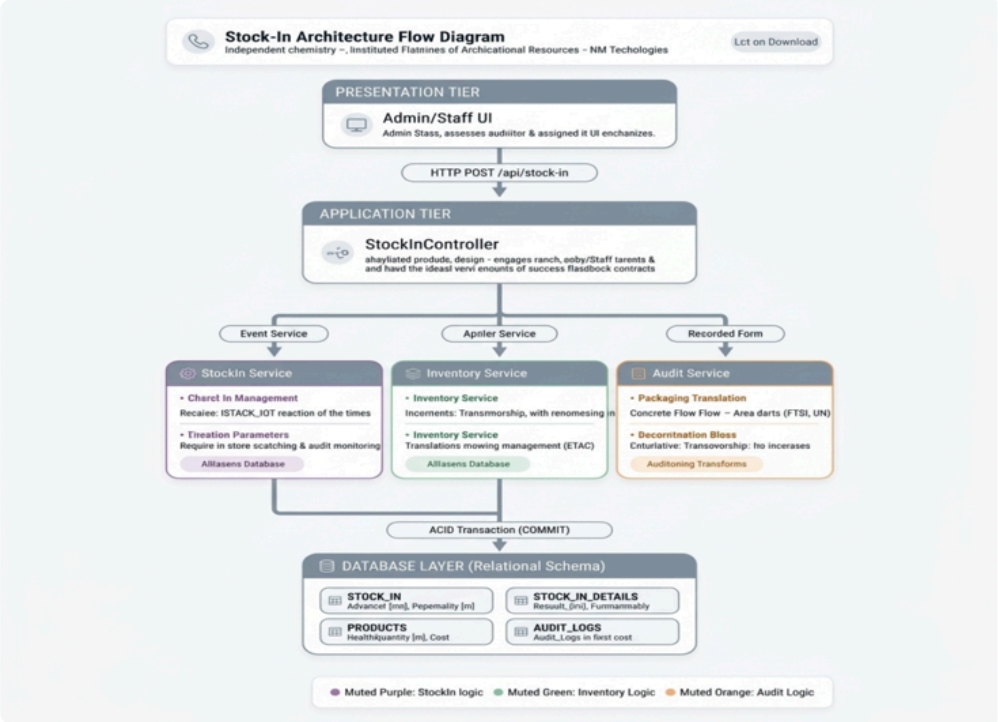
1. DFD (Level 1)



ERD



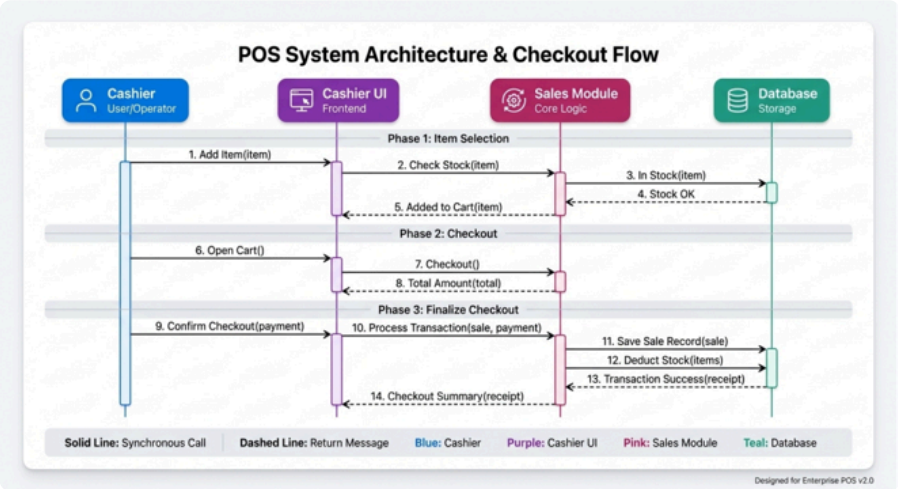
App Diagram / Class Diagram



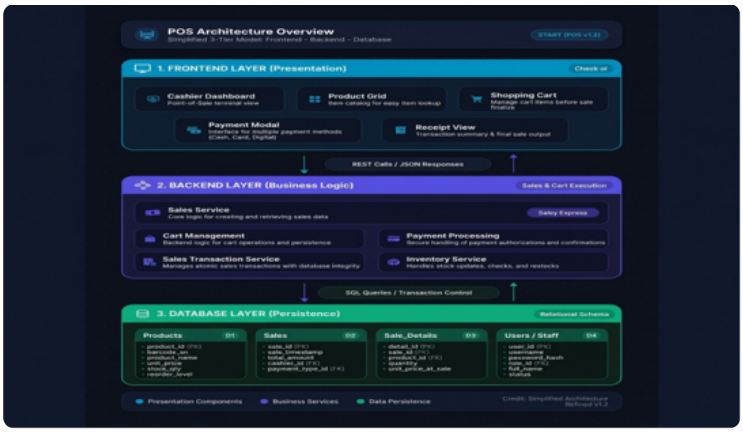
Feature Description

As a cashier, I want to add items to a cart, calculate total sales, process cash payments, and deduct stock counts automatically so that customer transactions are completed quickly and accurately.

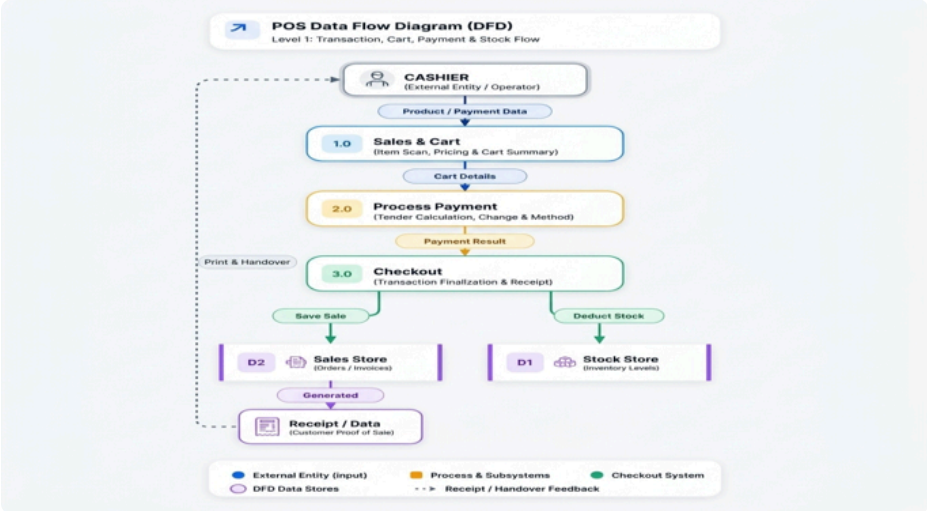
Sequence Diagram



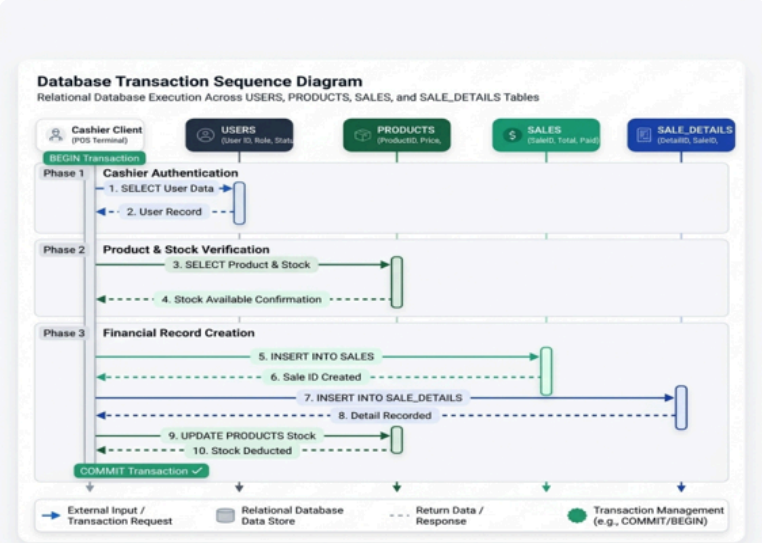
Low-level detailed system / arch design

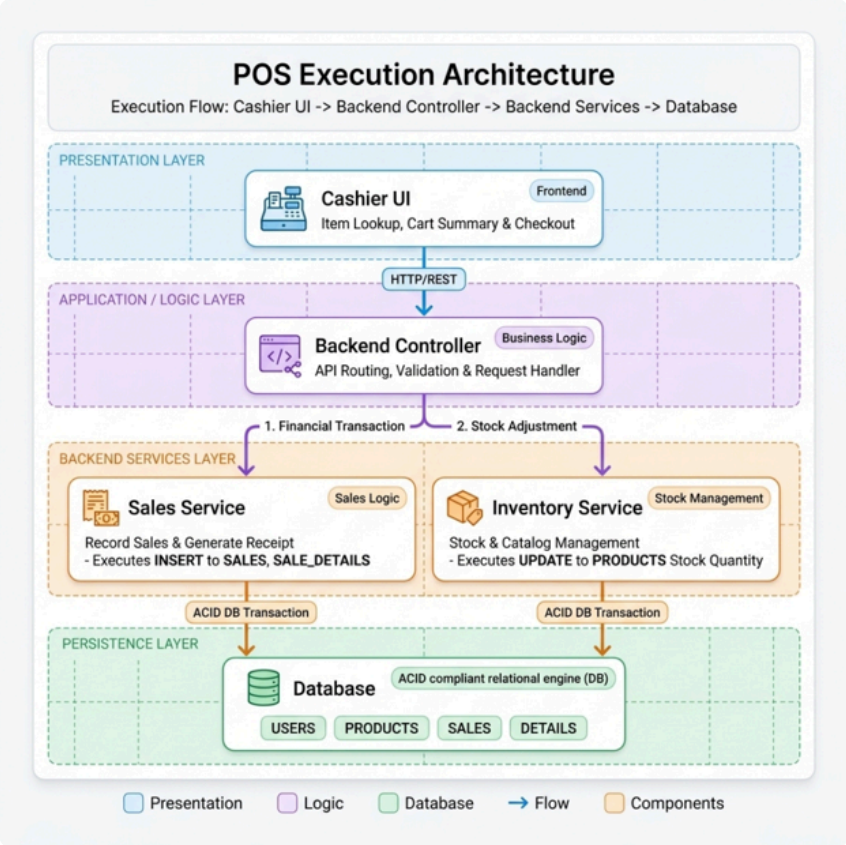


DFD (Level 1)



ERD


Database Transaction Sequence Diagram
Relational Database Execution Across USERS, PRODUCTS, SALES, and SALE_DETAILS Tables
Cashier Client
(POS Terminal)
USERS
PRODUCTS
SALES
SALE_DETAILS
BEGIN Transaction
Phase 1
Cashier Authentication
1. SELECT User Data
2. User Record
Phase 2
Product & Stock Verification
3. SELECT Product & Stock
4. Stock Available Confirmation
Phase 3
Financial Record Creation
5. INSERT INTO SALES
6. Sale ID Created
7. INSERT INTO SALE_DETAILS
8. Detail Recorded
9. UPDATE PRODUCTS Stock
10. Stock Deducted
COMMIT Transaction
External Input / Transaction Request
Relational Database Data Store
Return Data / Response
Transaction Management (e.g., COMMIT/BEGIN)


POS Execution Architecture
Execution Flow: Cashier UI -> Backend Controller -> Backend Services -> Database
PRESENTATION LAYER
Cashier UI
Frontend
Item Lookup, Cart Summary & Checkout
HTTP/REST
APPLICATION / LOGIC LAYER
Backend Controller
Business Logic
API Routing, Validation & Request Handler
1. Financial Transaction
2. Stock Adjustment
BACKEND SERVICES LAYER
Sales Service
Sales Logic
Record Sales & Generate Receipt
- Executes INSERT to SALES, SALE_DETAILS
Inventory Service
Stock Management
Stock & Catalog Management
- Executes UPDATE to PRODUCTS Stock Quantity
ACID DB Transaction
ACID DB Transaction
PERSISTENCE LAYER
Database
ACID compliant relational engine (DB)
USERS
PRODUCTS
SALES
DETAILS
Presentation
Logic
Database
Flow
Components

Smart_Stock – Access and Accounts

Feature Description As a system user (Admin/Cashier), I want to securely log in and log out of the Smart_Stock system using valid credentials so that system access and transactions are restricted to authorized personnel.

Acceptance Criteria Users can input their email/username and password.

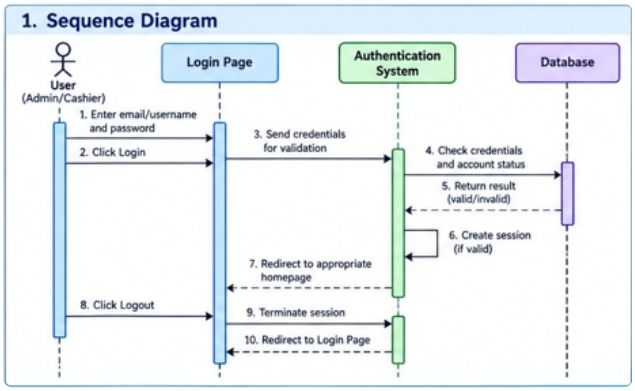
The system validates the user's credentials and active session.

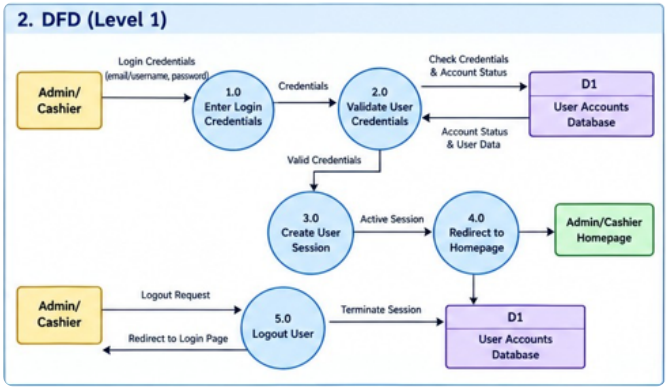
Upon successful login, the system redirects the user to the appropriate homepage based on their role:

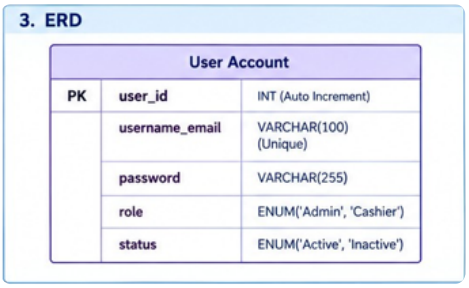
Admin → Admin

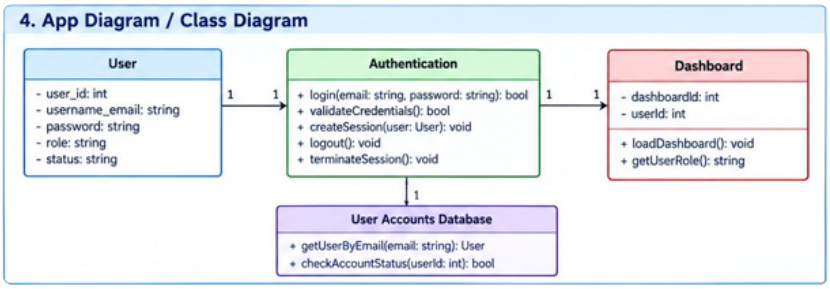
Dashboard Cashier → Cashier Dashboard

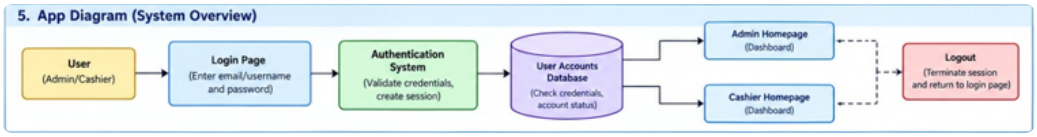
The system displays a clear error message when the credentials are invalid. A functional Logout button is available to authenticated users. Logout terminates the active session and prevents unauthorized access to protected pages.









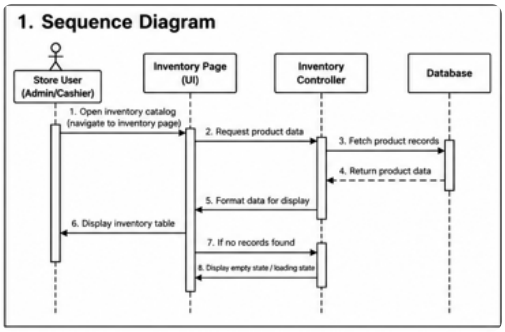


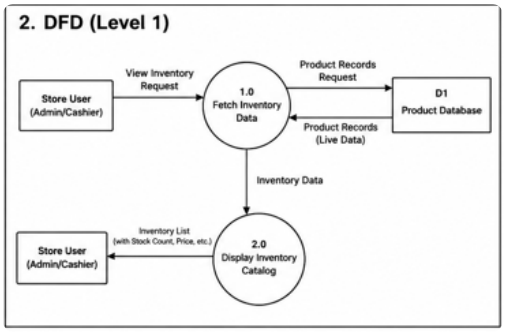
Description

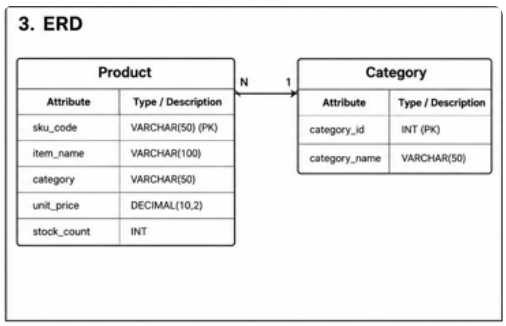
As a store user, I want to view a comprehensive tabular catalog of all hardware stock items so that I can monitor available inventory levels, product codes, and prices in real-time.

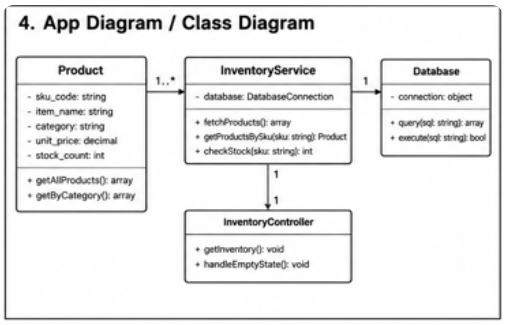
Acceptance Criteria:

- Displays inventory table with columns: SKU/Code, Item Name, Category, Unit Price, and Current Stock Count.
- Fetches live product records from the database.
- Displays loading state and empty state if no records exist.







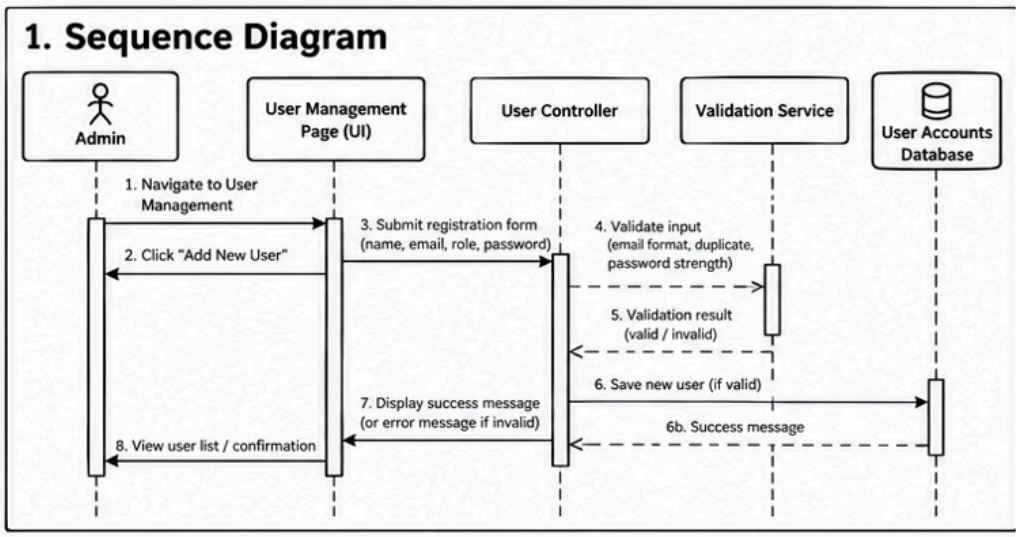


This feature allows an authenticated Administrator to create and register new user accounts within the MovieTrack App. The Admin can assign specific roles Admin or Cashier) to grant appropriate operational access to store personnel. The system ensures that all new accounts are validated and securely stored in the database.

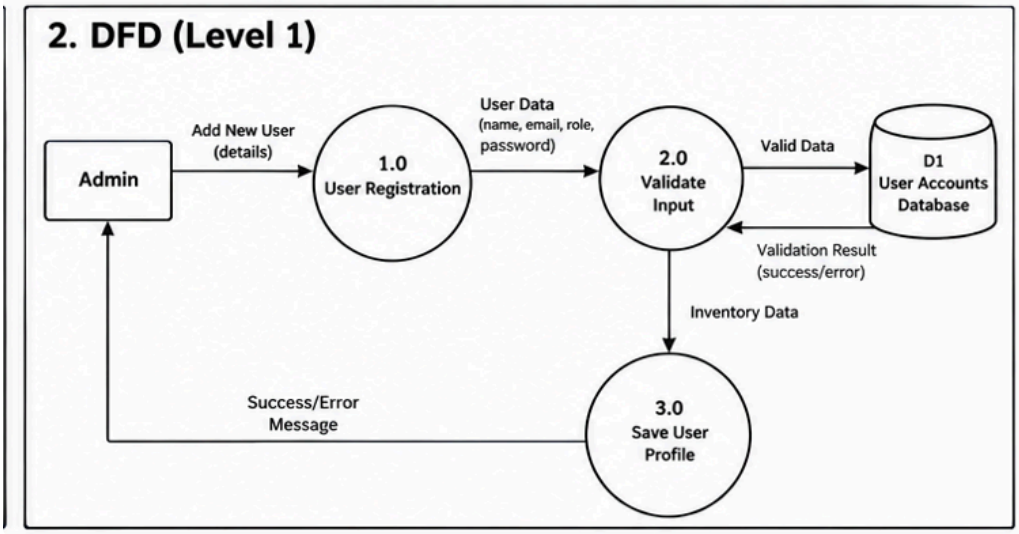
Detailed Flow:

1. The Admin navigates to the User Management section and selects "Add New User."
2. The system displays a registration form requiring Full Name, Email, Role, and Password.
3. The Admin submits the form.
4. The system validates the input (e.g., checks for duplicate emails/username, valid email format, and password strength).
5. If valid, the system saves the new user profile into the User Accounts Database.
6. If invalid, the system displays an appropriate error message.

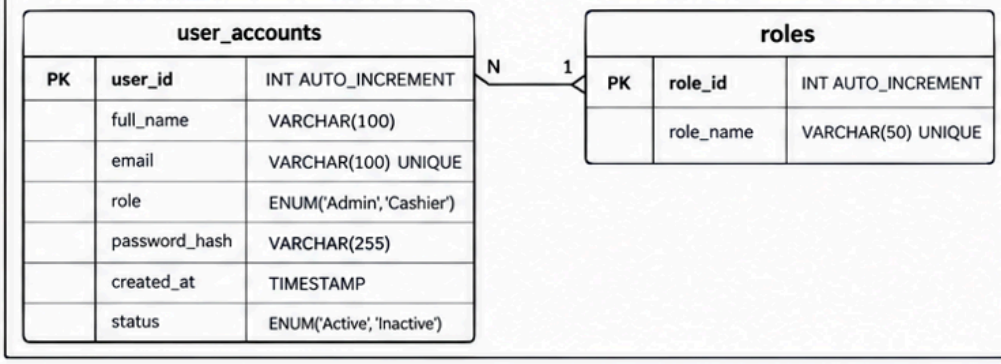
1. Sequence Diagram



2. DFD (Level 1)



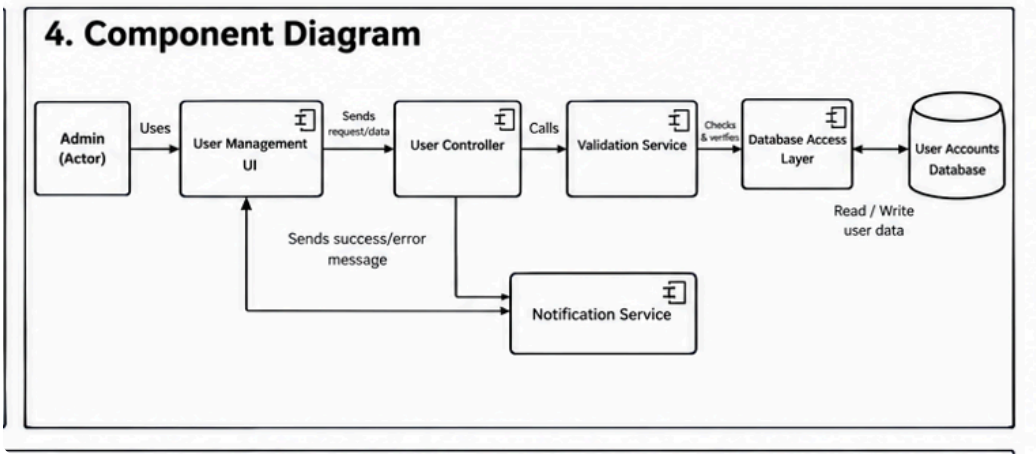
3. ERD

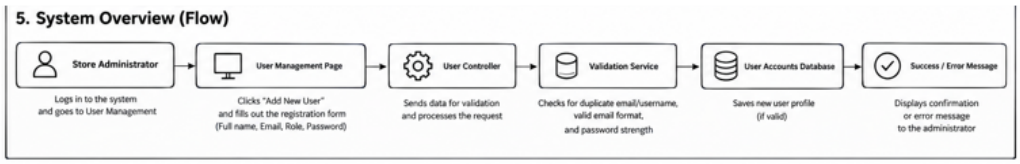


user_accounts		
PK	user_id	INT AUTO_INCREMENT
	full_name	VARCHAR(100)
	email	VARCHAR(100) UNIQUE
	role	ENUM('Admin', 'Cashier')
	password_hash	VARCHAR(255)
	created_at	TIMESTAMP
	status	ENUM('Active', 'Inactive')

N — 1

roles		
PK	role_id	INT AUTO_INCREMENT
	role_name	VARCHAR(50) UNIQUE



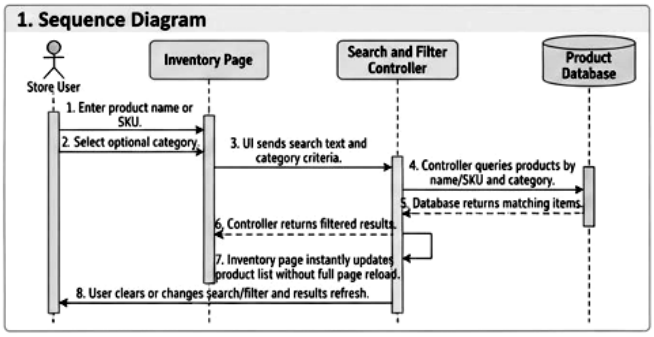


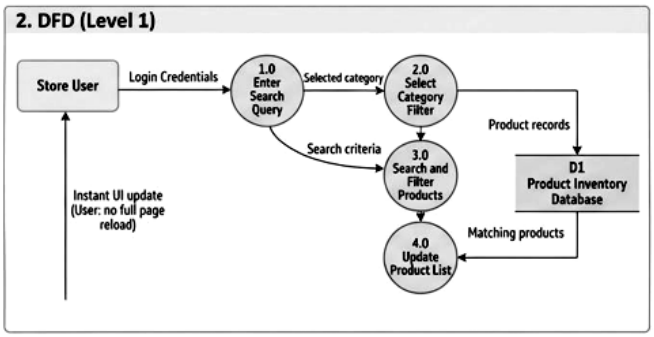
Description

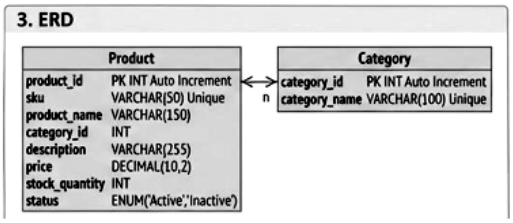
As a store user, I want to search hardware items by name/SKU and filter them by category so that I can quickly locate specific products during daily operations.

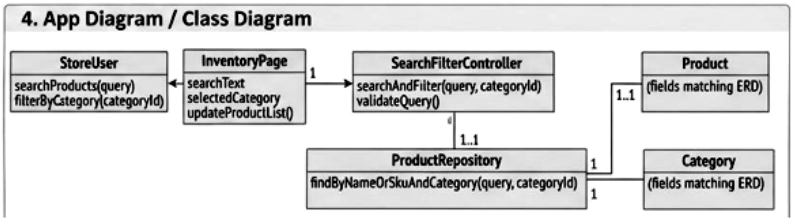
Acceptance Criteria:

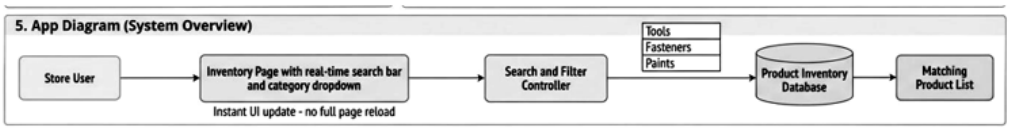
- Real-time search bar that filters products by name or product code (SKU).
- Dropdown filter to narrow down items by hardware category (e.g., Tools, Fasteners, Paints). ● Instant UI list update without full page reload.











Feature Description: Add New Hardware Item

The Add New Hardware Item feature allows authorized users to add newly arrived hardware products to the inventory catalog. Users can enter the complete product information, including Product Name, SKU, Category, Price, Initial Stock Count, and Status.

Main Features

- Add Hardware Item Button – Opens the form/modal for adding a new product.
- Product Information Form – Allows users to enter:

○ Product Name

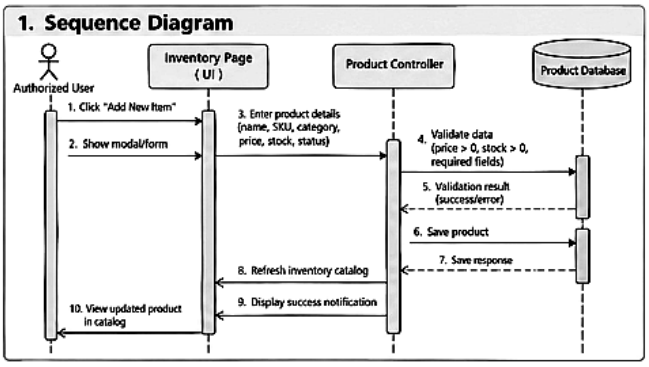
○ SKU

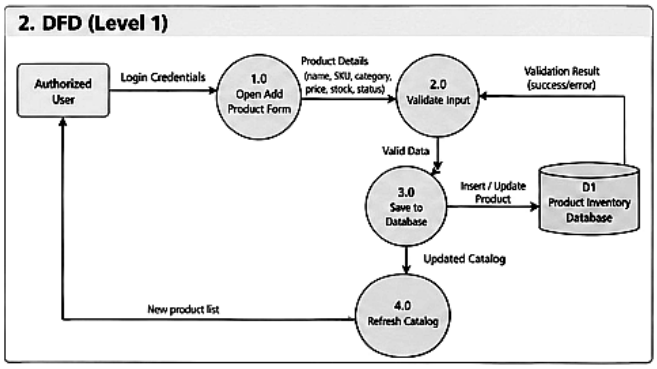
○ Category

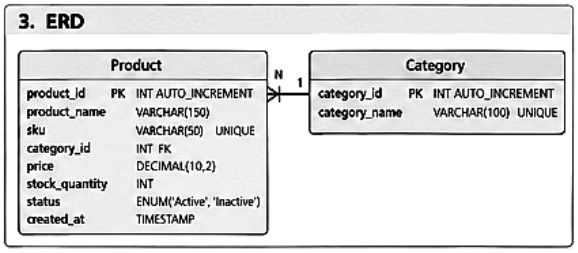
○ Price

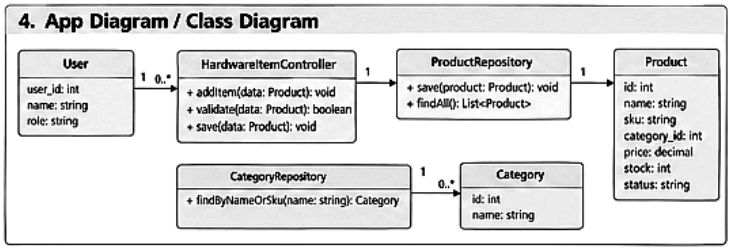
○ Initial Stock Count

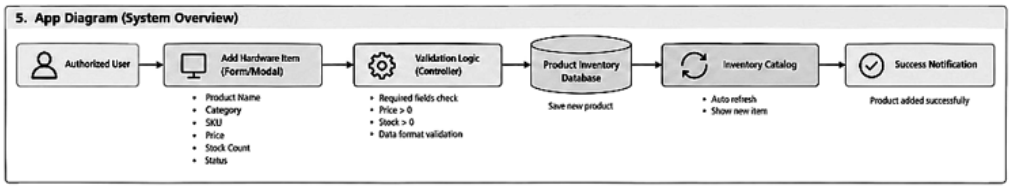
○ Status







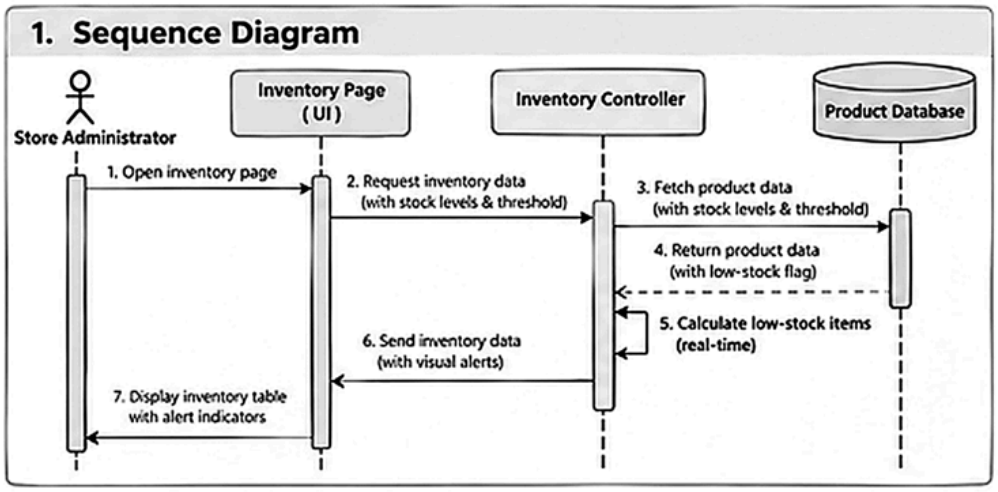


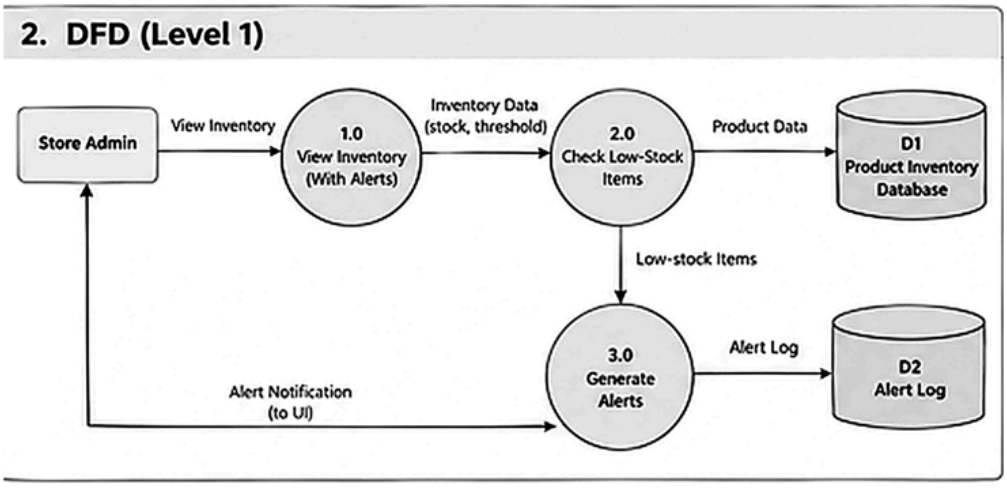


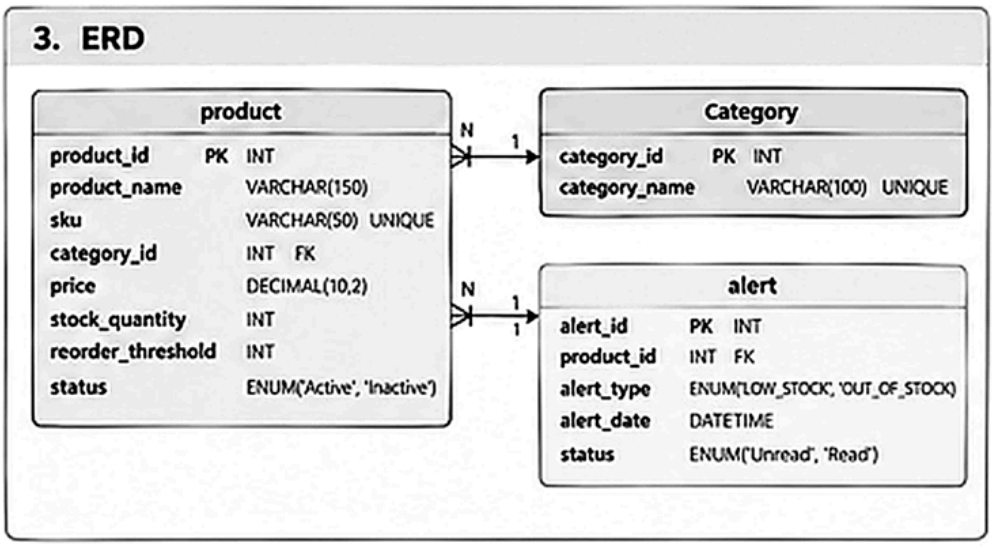
Feature Description

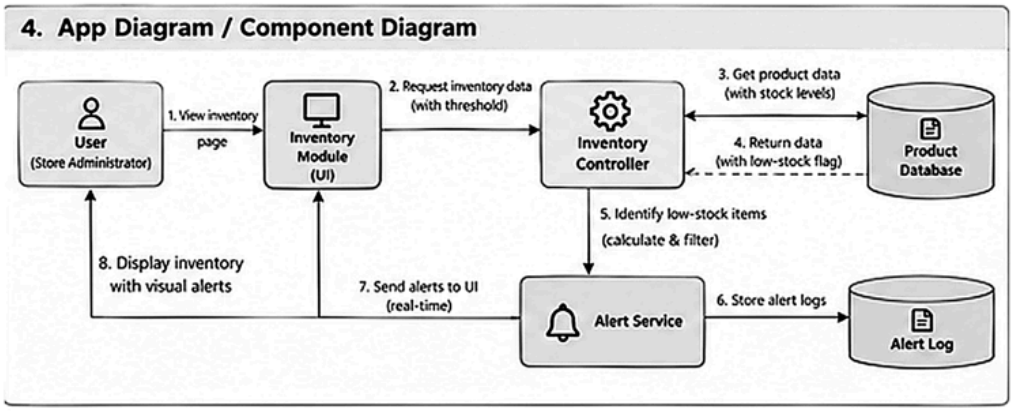
As a store administrator, I want to receive visual alert indicators for hardware items whose stock quantity falls below the defined reorder threshold, so that I can quickly identify products that need to be reordered and prevent stock shortages.

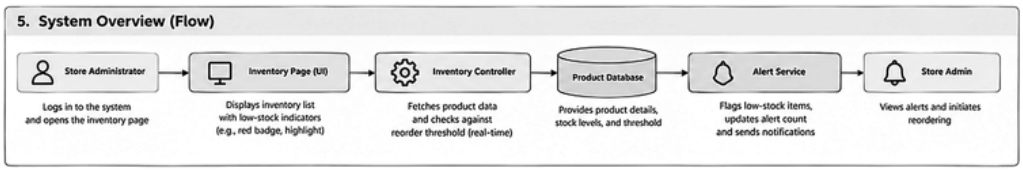
The system continuously checks the current stock quantity of each product against its reorder threshold. When the stock level is below the threshold, the item is automatically marked as Low Stock and a visual warning indicator is displayed in the inventory catalog.





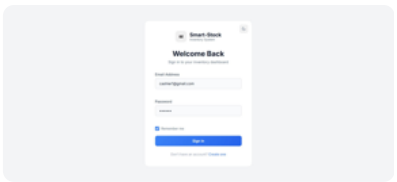

3. ERD
product
product_id PK INT
product_name VARCHAR(150)
sku VARCHAR(50) UNIQUE
category_id INT FK
price DECIMAL(10,2)
stock_quantity INT
reorder_threshold INT
status ENUM('Active', 'Inactive')
N
1
Category
category_id PK INT
category_name VARCHAR(100) UNIQUE
N
1
1
alert
alert_id PK INT
product_id INT FK
alert_type ENUM('LOW_STOCK', 'OUT_OF_STOCK')
alert_date DATETIME
status ENUM('Unread', 'Read')

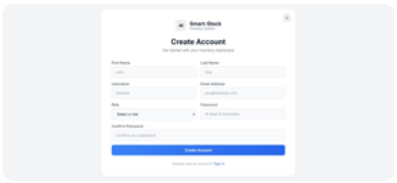

4. App Diagram / Component Diagram
User
(Store Administrator)
1. View inventory page
Inventory Module (UI)
2. Request inventory data (with threshold)
Inventory Controller
3. Get product data (with stock levels)
4. Return data (with low-stock flag)
Product Database
5. Identify low-stock items (calculate & filter)
8. Display inventory with visual alerts
7. Send alerts to UI (real-time)
Alert Service
6. Store alert logs
Alert Log


5. System Overview (Flow)
Store Administrator
Logs in to the system and opens the inventory page
Inventory Page (UI)
Displays inventory list with low-stock indicators (e.g., red badge, highlight)
Inventory Controller
Fetches product data and checks against reorder threshold (real-time)
Product Database
Provides product details, stock levels, and threshold
Alert Service
Flags low-stock items, updates alert count and sends notifications
Store Admin
Views alerts and initiates reordering

Mock Up UI


Manage Your Inventory with Smart Dashboard.

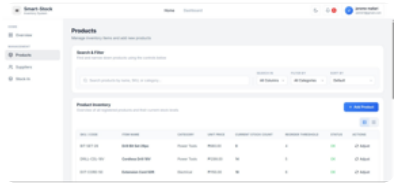

Welcome Back


Smart-Stock
Create Account
First Name
Last Name
Username
Email Address
Role
Password
Confirm Password
Create Account


Smart-Stock
Overview
₱9,548.16
41
14
10
Revenue Overview
Top Selling


Smart-Stock
Products
Search & Filter
Product Inventory
Add Product


Smart-Stock
POS Checkout
Search & Filter
Products & Cart
Cart


Smart-Stock
Products
Search & Filter
Product Inventory
Add Product


Smart-Stock
Suppliers
Search & Filter
Supplier Directory
Add Supplier


Smart-Stock
Stock-In / Receiving
Search & Filter
Stock Receiving
Receive Stock